



OLLSCOIL NA hÉIREANN MÁ NUAD
THE NATIONAL UNIVERSITY OF IRELAND MAYNOOTH

MATHEMATICAL PHYSICS

Year 3

Semester 1

2014 - 2015

Quantum Mechanics I
MP363

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Time allowed: $1\frac{1}{2}$ hours

Answer **two** questions

All questions carry equal marks

1. Let $|\phi_n\rangle$ be the n^{th} orthonormalised energy eigenstate of the hydrogen atom with energy $E_n = E_1/n^2$, where $E_1 = -13.6$ eV and $n = 1, 2, \dots$. We prepare a hydrogen atom so that it is initially in a state

$$|\psi\rangle = A \sum_{n=1}^{\infty} \frac{|\phi_n\rangle}{n},$$

where A is a positive real number.

- Determine the value of A such that $|\psi\rangle$ is normalised.
- Find the expectation value (in electron-volts) of the Hamiltonian for this state.
- A measurement of the atom's energy is made, and the value -0.85 eV is obtained. What state is the atom in immediately after getting this result?

The following sums may be useful:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \quad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

2. (a) Let M be an $n \times n$ complex matrix. Define how the adjoint operation $\dagger$ acts on such a matrix.
- (b) Explain what a self-adjoint operator is, and show that the three matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are self-adjoint.

- (c) Let $\mathcal{H}$ be the Hilbert space of 2×2 complex matrices with the inner product

$$\langle M|N\rangle = \frac{1}{2} \text{tr}(M^\dagger N),$$

where tr is the trace operation. Show that the three matrices in (b) are orthonormal under this inner product.

3. Consider the one-dimensional quantum-mechanical simple harmonic oscillator, where $\{|n\rangle | n = 0, 1, \dots\}$ are the orthonormalised eigenstates whose energies are $E_n = \hbar\omega(n + 1/2)$. The creation and annihilation operators are defined respectively as

$$a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(X - \frac{iP}{m\omega} \right), \quad a = \sqrt{\frac{m\omega}{2\hbar}} \left(X + \frac{iP}{m\omega} \right),$$

where X and P are the position and momentum operators. These act on the eigenstates via

$$a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle, \quad a |n\rangle = \sqrt{n} |n-1\rangle.$$

Suppose this oscillator is in the state $|\psi\rangle$ given by

$$|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle - \frac{i}{2} |1\rangle.$$

- If the energy of this state is measured, what are the possible results, and with what probabilities do they occur?
- Compute the expectation value of the Hamiltonian H .
- Compute the expectation values of the position operator X and the momentum operator P . (**Hint:** write these operators in terms of $a^\dagger$ and a .)